

Complexity Is Not an Edge

Complexity Is Not an Edge: An Empirical Study of Machine-Learning Arbitrage on Kalshi and Polymarket

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Abstract

We study cross-platform price discrepancies between **Kalshi** (CFTC-regulated event contracts) and **Polymarket** (on-chain prediction market) and ask whether increasing model complexity improves arbitrage detection. We build an end-to-end system that matches contracts via sentence embeddings plus a 10-rule structural quality filter, engineers 51 features, and trains four model tiers — regression baselines (Linear Regression, XGBoost), sequence models (GRU, LSTM), and PPO with and without an autoencoder anomaly filter — under one evaluation protocol. On 6,802 training rows / 1,673 test rows across 144 matched pairs at \$100 position size, **Linear Regression achieves a per-trade Sharpe of 0.501 with +15.0 bps per-trade alpha**, tied with XGBoost (0.499 / +14.9 bps), versus 0.473 / +14.3 bps for LSTM, 0.459 / +14.0 bps for GRU, +9.6 bps for PPO-Raw, and **+0.5 bps for PPO+autoencoder**. Every walk-forward window is profitable for every ML model and per-trade Sharpe rises from 0.31 to 0.53 over time. A pre-submission adversarial audit (`AUDIT_REPORT.md`) found that the original 80/20 row-index split bridged 142 of 144 pairs (embargo violation); a leakage-free pair-stratified retraining yields per-trade Sharpe 0.516 and +15.7 bps alpha — drift of +2.99%, well within the bootstrap CI. The per-trade edge is robust. The central empirical answer: **at this data scale, complexity is a liability, not an edge**. The alpha lives in the matching pipeline and the oil/commodities asset class, not in the models.

Keywords: prediction markets, arbitrage, market microstructure, XGBoost, LSTM, PPO, walk-forward validation, simplicity.

1. Introduction

Prediction markets allow traders to buy and sell contracts that pay \$1 if a specified real-world event occurs (e.g., “Will CPI inflation exceed 3.0% in May 2026?”) and \$0 otherwise. The equilibrium price is interpreted as the market-implied probability. Two large U.S.-accessible venues — **Kalshi** (CFTC-regulated, since 2021) and **Polymarket** (on-chain via Polygon, since 2020) — frequently list contracts that reference the *same* underlying event but trade at materially different prices. These cross-platform discrepancies can persist for hours or days, representing a potential statistical-arbitrage opportunity if they can be detected, matched correctly, and traded before they close.

The central research question is:

Does increasing model complexity improve arbitrage detection in cross-platform prediction markets, and if so, when is that complexity justified?

Two distinct audiences motivate this work. *Academically*, a January 2026 working paper (arXiv 2601.07131) argues that feature engineering grounded in market-microstructure theory consistently beats deep learning for investor-flow prediction. Prediction-market arbitrage is a clean testbed because contracts have finite lifetimes, bounded payoffs, and deterministic settlement. *Practically*, prediction markets are growing rapidly: Polymarket processed over \$1B in volume during the 2024 election cycle. Understanding which modeling approaches genuinely add value is a relevant question for market participants.

Background and prior work. A binary event contract on platform P at time t has price $p_P(t) \in [0, 1]$; for two platforms listing the same event, the spread $s(t) = p_A(t) - p_B(t)$ converges to zero at resolution modulo fees and basis risk. We predict the one-step change $\Delta s(t) = s(t+1) - s(t)$. Manski (2006) and Wolfers & Zitzewitz (2004) established that prediction-market prices are reasonably efficient probability forecasts, but cross-platform dispersion is understudied. Bürgi, Deng & Whelan (2026) document a clear favorite–longshot bias on Kalshi (low-price contracts win less often than required to break even, while high-price contracts yield small positive returns) — a structural pricing pattern that, if asymmetric across platforms, becomes a source of cross-platform spread. Standard market-microstructure estimators — Amihud’s (2002) illiquidity ratio, Corwin & Schultz’s (2012) high-low bid–ask spread, Kyle’s (1985) λ , and Roll’s (1984) implied spread — are well established in equity markets but, to our knowledge, have not been systematically applied to prediction-market spread prediction. A persistent finding in applied ML is that on tabular data with moderate sample size, gradient-boosted trees match or beat deep neural networks (Grinsztajn et al. 2022); our results are consistent with this pattern.

Contributions. (1) A functioning end-to-end arbitrage system, from data ingestion through live paper-trading on BU SCC, monitoring up to 11,582 matched pairs at peak. (2) A complexity-vs-performance benchmark across four model tiers under an identical evaluation protocol. (3) An 11-window walk-forward validation showing the edge is stable and *improving* over time. (4) Proposal-mandated parameter searches over lookback windows and minimum-spread thresholds. (5) An honest Sharpe accounting that survived a pre-submission adversarial audit. (6) A negative result on PPO that we report transparently.

2. Methodology

2.1 Data Sources and Matching

Kalshi exposes a public REST API (`api.elections.kalshi.com/trade-api/v2`) requiring no authentication. We fetch hourly OHLCV candlesticks (`period_interval=60`); Kalshi splits market history at a roughly three-month cutoff, so we query `/historical/cutoff` to determine which endpoint applies. **Polymarket** exposes three separate APIs: Gamma (metadata), CLOB (orderbook/prices for active markets), and the Data API (fills/trades for resolved markets). Polymarket uses opaque numeric token IDs, so we look up `clobTokenIds` via Gamma and key CLOB queries by them. A non-obvious gotcha: Gamma’s singular `condition_id=` query parameter returns *unrelated random markets*, while the plural `condition_ids=` returns the exact match. The `/prices-history` endpoint returns empty data for resolved markets, forcing reconstruction from trade records.

Matching pipeline. We embed concatenated title + description with `sentence-transformers/all-MiniLM-L6-v2` (384-d) and compute cosine similarity via normalized matrix multiplication (~ 80 s vs ~ 6.6 h for an $O(N \cdot M)$ keyword pre-filter — a $300\times$ speedup). Semantic similarity alone is insufficient (“NBA Finals winner” embeds close to “NBA MVP”), so we layer **10 structural quality-filter rules**: sports wins-vs-champion mismatch, Fed meeting-month mismatch, cabinet confirmation-vs-nomination, state-specific vs national commodities, exact CPI vs PCE vs core-CPI, ± 2 -day expiry window, exact strike comparison for futures, numeric-threshold exactness (“\$50/bbl” vs “\$55/bbl”), event-key disambiguation (“Game 4” vs “Game 5”), and a category-consistency guard. The filter rejected **140 of 615 pairs (22.8%)**. Empirical impact: at the first large-scale backtest, Linear Regression P&L went from $-\$5.28$ to $+\$5.45$ — a $+\$10.73$ swing purely from removing structurally-bad matches, with no model changes. Data quality is more important than model choice at this scale.

2.2 Feature Engineering

We engineer 59 raw features and use 51 numeric features after dropping NaN/zero-variance columns. Five groups: 18 raw aligned per-platform OHLCV features (e.g., `kalshi_open`, `polymarket_volume`); 6 cross-platform basics (`spread`, `mid_price`, `dollar_volume_ratio`); 9 rolling/momentum features (`spread_momentum_6`, `spread_zscore`); 13 classical microstructure features (`amihud_illiquidity`, `corwin_schultz_spread`, `kyle_lambda`, `roll_spread`, `bekker_parkinson_vol`) computed per-platform and differenced across platforms; and 13 prediction-market-specific features (`favorite_longshot_bias`, `near_expiry_indicator`). The rationale for differenced microstructure features is that *relative* liquidity should predict which side converges. These features are neutral-to-slightly-positive on the historical dataset (47 bars/pair on average is too few for stable rolling-window estimators) and are expected to become informative as live data accumulates.

2.3 Model Tiers

All models share the same train/test split (time-ordered 80/20, no shuffling), the same target ($\Delta s(t)$), and the same feature set.

- **Tier 0 — Naive baselines:** *Naive-closes* (predict spread fully closes by resolution); *Volume-higher-wins* (higher-volume platform is correct).
- **Tier 1 — Regression:** *Linear Regression* (scikit-learn); *XGBoost*, hyperparameter-searched over $\text{depth} \in \{3, 5, 7, 9\} \times \text{learning-rate} \in \{0.01, 0.05, 0.1, 0.3\} \times \text{n_estimators} \in \{100, 300, 500\}$ (48 configurations).
- **Tier 2 — Sequence models:** *GRU* (64 hidden units, 1 layer, 24-bar lookback, Adam $\text{lr}=1\text{e-}3$, early stopping); *LSTM* (same architecture, LSTMCell). We also attempted *TFT* (PyTorch Forecasting, `hidden_size=8`, 3-quantile QuantileLoss) with small-data hyperparameters; it did not converge (§3.4).
- **Tier 3 — Reinforcement learning:** *PPO-Raw* (PPO on 51-d feature vectors, 3-action space {buy-spread, sell-spread, hold}, mark-to-market reward; `stable-baselines3`); *PPO-Filtered* (same PPO, but a 3-layer symmetric autoencoder with `bottleneck=8` pre-filters observations, gating PPO to anomaly windows).

2.4 Evaluation Protocol

Five regimes provide independent views. **(a) Single-split backtest:** chronological 80/20, reporting RMSE, MAE, directional accuracy, simulated P&L at the 2 pp signal threshold, win rate, and per-trade Sharpe — the headline numbers. **(b) Walk-forward backtest:** concatenate train + test for maximum time coverage, split into 12 equal-time windows, and use an expanding-window protocol where window i trains on data from windows $\{0, \dots, i-1\}$ and tests on window i (window 0 has no training set, leaving 11 evaluation windows). If the edge

is stable, we expect positive P&L across all windows; if it is improving with more data, per-trade Sharpe should trend upward. **(c) Per-category breakdown:** stratify test rows by contract category using deterministic rules over Kalshi tickers and Polymarket slugs. **(d) Data-scaling curve:** 50, 100, 250, 500, 1000, 2000 bars/pair, plotting P&L vs training size — does “simple wins” hold across scales or is it an artifact of small data? **(e) Live paper trading** on BU SCC with a 15-minute trade cycle, 3-hour discovery batch, and 6-hour retrain batch, with redundant fallback workflows on GitHub Actions in case SCC undergoes scheduled maintenance. Tier-1 models train in <15 s on a CPU; Tier-2 in ~3 min/epoch with early stopping at 10–15 epochs; Tier-3 in 20–40 min for PPO convergence. Live deployment uses an LR + XGBoost equal-weight ensemble with a strict concordance filter (skip trades when models disagree on sign), and content-addressed pair IDs (e.g., `kxwti26apr08t10799-0x43d5953d`) derived deterministically from the normalized Kalshi ticker and Polymarket token ID — fixing an earlier bug where three code paths disagreed on what a `live_NNNN` pair ID meant, causing 25 positions to track the wrong markets.

2.5 Infrastructure Lessons

Three non-obvious infrastructure bugs materially shaped the project. **Kalshi** `/events` silently returned HTTP 429 on roughly 40% of calls, dropping entire commodity series; fixing with exponential backoff and a 250 ms per-series pace took commodity pair count from 65 → 506. **Polymarket pagination defaulted too shallow** at 5,000 markets; WTI markets sat at offset 15,305+, completely invisible until we raised `max_pages` from 10 to 60. **Gamma’s** `condition_id` (singular) returned unrelated random markets rather than an error; the fix to `condition_ids=` (plural) was a one-character change but cost several days of debugging. The common pattern: APIs that fail silently in ways that *look like* model problems. Infrastructure monitoring must come before model tuning.

3. Results

3.1 Headline Model Comparison

Table 1 reports the single-split backtest at a 2 pp signal threshold for trade entry on the full 1,673-row test set. All numbers are sourced verbatim from `experiments/results/canonical/headline.json` (Phase 17-01 canonical regenerator under seed=42, threshold=0.02, position_size=\$100). The per-trade Sharpe and per-trade alpha (in basis points) lead the comparison; cumulative dollar P&L follows.

Table 1: Single-split backtest (canonical, 2 pp signal threshold, 1,673-row test set).

Tier	Model	RMSE	Dir. Acc.	Sharpe (per-trade)	Alpha (bps/trade)	P&L (\$100 pos)	Win Rate	# trades
0	Naive (closes)	0.499	47.7%	0.125	+4.0	+\$58.12	47.9%	1460
0	Volume (higher wins)	0.457	47.7%	0.131	+4.2	+\$59.81	48.1%	1440
1	Linear Regression	0.306	56.9%	0.501	+15.0	+\$232.67	57.8%	1549
1	XGBoost (depth 3)	0.290	56.6%	0.499	+14.9	+\$232.83	57.4%	1559
2	LSTM	0.291	65.5%	0.473	+14.3	+\$221.84	56.5%	1547
2	GRU	0.293	64.3%	0.459	+14.0	+\$212.50	55.8%	1517
2	TFT†	0.326	51.7%	0.155	+5.5	+\$6.57	37.5%	120
3	PPO-Raw	0.319	60.5%	0.306	+9.6	+\$158.15	52.2%	1656
3	PPO + Autoencoder	0.327	27.2%	0.014	+0.5	+\$4.61	43.2%	899

†TFT did not converge at N=6,802 (30-epoch budget, hidden_size=8); see §3.4.

The ranking is unambiguous: **Tier 1 > Tier 2 > Tier 3**. The Tier-0 → Tier-1 gap is large ($\approx 4\times$ in per-trade alpha: 4 bps → 15 bps); the Tier-1 → Tier-2 gap is 0.7–1.0 bps; PPO+autoencoder collapses to essentially zero edge. Linear Regression and XGBoost are essentially tied (Sharpe 0.501 vs 0.499; +15.0 vs +14.9 bps); within Tier 1, there is no statistically defensible reason to prefer one over the other on this dataset. Bootstrap 95% confidence intervals on RMSE (Fig. 7) overlap heavily across LR/XGBoost/GRU/LSTM even though P&L separates cleanly — the gap is driven by *directional accuracy* and *trade selection*, not raw regression error.

3.2 Walk-Forward Validation

We evaluate each model across 11 expanding-window splits spanning Jan–Apr 2026. Per-window P&L (Fig. 1) is positive in **every window for every ML model** — $11/11 \times 4/4$, zero losing windows. This is strong evidence the edge is not a lucky train/test split. Per-trade Sharpe trends upward across windows: LR rises from 0.307 in window 1 (357 training rows) to 0.530 in window 10 (8,012 training rows), a 73% improvement; XGBoost shows the same pattern (0.383 → 0.540). Median per-trade Sharpe across windows is 0.408 (LR), 0.424

(XGBoost), 0.410 (GRU), 0.405 (LSTM) — all four ML models converge to near-identical median performance. The economically meaningful comparison is *any ML model* (median $\sim \$75/\text{window}$) vs *no ML* (median $\sim \$25/\text{window}$), not XGBoost vs GRU.

3.3 Parameter Searches (Proposal Experiments)

XGBoost hyperparameter sweep (48 configurations). The best configuration is `depth=3, lr=0.01, n_estimators=100` (RMSE 0.288, +\$209.70 P&L). All top-10 configurations use $\text{depth} \in \{3, 5, 7\}$; depth-9 trees lost P&L, evidence of over-fitting at 6,802 training rows. Within a single model family, “simple wins” recurs: the best XGBoost is essentially an ensemble of decision stumps.

Lookback window sweep (Proposal Experiment 2). We sweep $\text{lookback} \in \{2, 6, 12, 18\}$ bars (8, 24, 48, 72 hours of hourly data) for both GRU and LSTM, holding all other hyperparameters fixed (Fig. 8). For GRU, P&L declines monotonically from \$+226.42 at 2 bars to \$+191.90 at 18 bars (−15%); per-trade Sharpe falls from 0.489 to 0.400. LSTM is flatter (best at 6 bars: \$+220.37; worst at 18 bars: \$+209.42) but trends in the same direction. **Longer lookbacks do not help, and at 72 h they meaningfully hurt.** With pairs averaging 47 bars total, an 18-bar window covers nearly 40% of contract lifetime and is dominated by stale pre-active-phase information. Sequence models cannot exploit longer history at this scale.

Minimum-spread threshold sweep (Proposal Experiment 3). We sweep the decision threshold $\in \{0.00, 0.02, 0.05, 0.10\}$ across all eight models (Fig. 9). For ML models, P&L peaks at threshold 0.00–0.02 and declines at higher thresholds (XGBoost loses \$36 going from 0.00 $\rightarrow$ 0.10). However, *per-trade Sharpe rises monotonically* with threshold for every ML model (XGBoost: 0.498 $\rightarrow$ 0.620, +24%): higher thresholds discard low-magnitude trades faster than they sacrifice profitable ones. Naive baselines show the *opposite* P&L pattern — they have no predictive content, so filtering to large-spread trades is the only thing that helps them. We use threshold 0.02 as the fee-consistent baseline throughout. The takeaway: threshold selection is a second-order tuning knob compared to choice of model tier.

Transaction-cost sensitivity (Fig. 4). The system is profitable at every fee level from 0 pp (gross +\$245) to a worst-case 7 pp (+\$135). Net P&L compresses roughly linearly ($\sim \$18$ per pp), but per-trade Sharpe is remarkably flat — fees filter out lower-confidence trades, partially offsetting the cost. Model rankings are invariant across the 0–10 pp fee range. In practice, Kalshi maker orders pay \$0 fee and Polymarket charges only ~ 1 pp in Polygon gas, so our 2 pp simulation is *conservative*.

Feature ablation (pre-registered). To guard against p-hacking, we pre-registered a Leave-One-Group-Out ablation protocol at `.planning/ablation_protocol.md` before any experiment script was written, with five non-overlapping feature groups (raw OHLCV, cross-platform basics, rolling/momentum, classical microstructure, prediction-market-specific).

Across 12 LOGO configurations (LR + XGBoost $\times$ 6 group conditions) on a 1,021-row ablation-holdout, no group meets the pre-registered load-bearing threshold (95% CI fully below zero AND $|\Delta P\&L| > \$10$ for both models). At this holdout size, paired-bootstrap CIs are too wide to detect group-level effects smaller than $\sim \$10$. We therefore conservatively retain all 51 features and defer feature selection to a re-run at 250+ bars/pair where statistical power becomes sufficient. The final 1,673-row test set was not evaluated on any reduced-feature variant.

3.4 Other Findings (Compressed)

Per-category stratification. Linear Regression wins 5 of 7 sufficiently-traded categories; XGBoost wins crypto by \$6 (its tree-based splits capture nonlinear volatility regimes that LR cannot). On the live April-11 dataset, oil near-expiry contracts dominated with **76.5% win rate and +\$0.41/trade** versus essentially zero for sports and politics — convergence is mechanical when WTI-futures expiry dates resolve on observable market prices. The alpha is in the asset class as much as in the model.

Data-scaling curve. P&L plateaus by 100 bars/pair because train.parquet caps at 6,802 rows / 144 pairs (max 141 bars/pair); slices at 250+ bars/pair are identical. Within the measured range, the ranking XGBoost > LR > GRU > LSTM is **invariant** across a $5\times$ growth in training data — refuting the hypothesis that sequence models would overtake regression with more data, at least at scales we can reach.

SHAP feature importance. `polymarket_vwap` dominates with mean $|\text{SHAP}| \approx 0.14$, twice the next feature, suggesting Polymarket may be the slower-reacting price-discovery side (Fig. 5).

Live ensemble. The production system uses an LR + XGBoost equal-weight ensemble with a strict concordance filter. An 11-point LR-weight sweep ($0.0 \rightarrow 1.0$) shows P&L spans only \$4.68 — weight choice is immaterial. The concordance filter rejects 4.80% of trades and improves per-trade Sharpe from 0.437 (unfiltered) to 0.455 (filtered), with a measurable cost: rejected trades would have netted +\$1.95.

TFT non-convergence. TFT did not converge at $N=6,802$ with the small-data hyperparameters (avg RMSE 0.3262 vs GRU 0.2928, only 120 trades vs 1,517). VSN attention entropy (2.656) confirms the attention is healthy and not degenerate (Fig. 10) — the bottleneck is data volume, not architecture correctness. At 1,000+ bars/pair a larger TFT (hidden_size=32+) becomes justifiable.

3.5 Adversarial Audit and Honest Sharpe Accounting

Per-trade Sharpe is the load-bearing headline. A pre-submission adversarial audit (Phase 18, `AUDIT_REPORT.md`) ran six independent kill-or-confirm checks against every quantitative claim in this paper: Tier 1 (headline Sharpe), Tier 2 (leakage / look-ahead), Tier 3 (cost

realism), Tier 4 (survivorship / selection), Tier 5 (paper number-by-number trace), Tier 6 (live-vs-backtest honesty). Tier 2 flagged that the canonical 80/20 row-index split bridged **142 of 144 pairs** between train and test, with a typical gap of just 4 hours between train end and test start for the same `pair_id` — an embargo violation: the same underlying market events inform both training and test losses. We rebuilt the split as a pair-stratified 80/20 (115 train pairs / 29 test pairs, seed=42, with `train_pairs.isdisjoint(test_pairs)` enforced by construction) and retrained Linear Regression and XGBoost on the leakage-free split.

Per-trade Sharpe drift was only +2.99% (0.501 → 0.516) and per-trade alpha drifted +4.5% (+15.0 → +15.7 bps) — both well within the 10,000-resample bootstrap CI. The per-trade edge is robust to leakage correction. The per-pair *annualized* Sharpe moved more dramatically because the Bailey–López de Prado (2012) cross-pair-correlation correction $n_{\text{eff}} = N / (1 + (N - 1) \cdot \text{avg_corr})$ flips between regimes: the leaky sample has avg pairwise correlation +0.042 (BLdP haircut active, compressing 0.78 → 0.30); the purged sample has avg correlation −0.199, which short-circuits the correction (`avg_corr ≤ 0` returns naive). We therefore demote the per-pair number to a regime-dependent secondary statistic with a 95% bootstrap CI of [0.700, 1.067] on the purged sample. An earlier draft cited a per-pair annualized Sharpe of ≈ 3.2 in the abstract; the audit found this value had no derivation path in the codebase and replaced it with the per-trade headline above. Three Tiers (1, the canonical-split portion of 2, and 3) returned non-PASS verdicts and were corrected in this revision; Tiers 4, 5, and 6 returned PASS without paper changes. The full audit report is at `AUDIT_REPORT.md`.

For comparison, mid-frequency stat-arb strategies typically target 1–5 bps/trade with millions of trades; our 15 bps/trade reflects the larger-than-typical mispricings in immature cross-platform prediction markets, and the trade-count ceiling ($\sim 1,500$ in the test set) caps total scalability. PPO+autoencoder at +0.5 bps/trade has essentially no edge; PPO-Raw at +9.6 bps is mildly profitable but ~ 5 bps below the regression baselines. PPO+autoencoder *destroys* roughly 9 bps of edge that PPO-Raw alone would have captured — a direct empirical answer to the professor’s question of whether RL plus anomaly detection improves on simpler regression: no, it actively hurts. We verified the reward function, environment transitions, and action space; the autoencoder simply flags normal market behavior as anomalous because it was trained on all spreads without a clean “normal regime” prior. PPO then trades disproportionately in high-volatility windows where predictions are least reliable.

3.6 Live Paper Trading

The live system was deployed on BU SCC on April 11, 2026. Through April 22, it accumulated **10,154 closed positions** across 8,421 pairs, with 100% match rate to entry bars (acceptance gate $\geq 80\%$ passed without margin). Live realized P&L was +\$1.53 versus −\$1.53 in shadow simulation — an exact directional anti-correlation traced to a model-semantics mismatch (the regression models predict next-bar spread change as a mean-reversion signal; the live strategy

enters short-spread on large spreads, opposite to `sign(prediction)`). The strategy's edge derives from spread-magnitude thresholding rather than directional alignment with predictions. Crypto sub-category P&L flipped sign (positive at the 3-day snapshot, negative at 8 days, $-\$19.87$ across 915 trades) — a documented stationarity caveat for per-category claims.

WTI oil contracts were absent from the April 14–22 live window (zero closed positions), so the backtest oil edge could not be live-validated within the original submission. Between submission and final draft, we diagnosed the root cause: `KALSHI_DISCOVERY_CATEGORIES` in `src/live/market_discovery.py` omitted `"Commodities"`, and Kalshi had migrated daily WTI/Brent/grain/metal series into that category. A single-line fix (commit `38d7970`), combined with a category-classifier extension and a 200-slot commodity reservation in the live collector, unblocked the pipeline. **Post-fix 12-hour window (April 24): 1,224 closed commodity positions** (`KXBRENTW`=486, `KXWTI`=409, `KXWTIW`=213, `KXBRENTMON`=76, plus four smaller series), with aggregate P&L $+\$1.96$ and 36.0% win rate. This is a *proof-of-life* result, not a robust live edge measurement — the 12-hour window spans one regime, the per-trade economics ($\sim \$0.0016$) are dramatically below the backtest oil near-expiry edge of $+\$0.41/\text{trade}$, and the live cohort includes full-contract-lifecycle positions rather than only the near-expiry subset the backtest measured. The lower live win rate is therefore expected rather than contradictory; a two-proportion z-test against the 76.5% backtest figure yields $z = -10.76$, $p \approx 5 \times 10^{-27}$, but the audit (Tier 6) confirms the comparison is between non-equivalent cohorts. Future work can now measure live oil-edge stability over longer windows.

4. Conclusions

We built an end-to-end cross-platform prediction-market arbitrage system with four model tiers and evaluated it across five independent regimes. The central answer to our research question is clear:

At this data scale, increasing model complexity does not improve arbitrage detection. The simplest models win.

Specifically: **Tier 1** (LR $+15.0$ bps/trade, Sharpe 0.501; XGBoost $+14.9$ bps, Sharpe 0.499) beats **Tier 2** (LSTM $+14.3$ bps, GRU $+14.0$ bps) by 0.7–1.0 bps and 5–10% absolute P&L; **Tier 3** (PPO+autoencoder $+0.5$ bps) is essentially zero alpha. This ordering holds across single-split, walk-forward, and data-scaling evaluations. Every walk-forward window is profitable for every ML model, and per-trade Sharpe rises with more training data — the edge is stable and improving. The system is robust to transaction costs at every fee level from 0 to 7 pp.

Four reasons appear to operate jointly. **Sample size:** with 47 bars/pair on average and 6,802 total training rows, GRU/LSTM operate well outside the 200–1000-timestep regime where sequence models excel; they are data-starved by construction, and the lookback ablation (§3.3) confirms longer sequences actively hurt rather than help on this dataset. **Signal-to-noise:** the one-step spread-change target has high variance relative to signal, and linear models are more robust in high-noise regimes because they have fewer parameters to over-fit. **Feature engineering carries the signal:** our 51 features include `spread`, `polymarket_vwap`, `dollar_volume_ratio`, and 13 microstructure features — they encode most exploitable signal directly, so the model only needs to linearly combine them, and deep models waste capacity re-discovering what the features already expose. **Overfitting on structural regularities:** the deepest XGBoost configurations (depth 9, `n_estimators` 500) had higher training accuracy but *lower* test P&L; the same phenomenon appears in sequence models, which memorize per-pair idiosyncrasies that do not generalize. This pattern matches the broader tabular-ML literature (Grinsztajn et al. 2022) and the January 2026 finding on investor-flow prediction.

We expect the picture to evolve as the live auto-retrain system accumulates data. By 250 bars/pair (currently a few weeks away), GRU should close the gap with XGBoost; by 500 bars/pair, sequence models may surpass XGBoost on oil and crypto categories where lead-lag effects dominate. By 1,000 bars/pair, a larger TFT configuration becomes justifiable. This is a falsifiable prediction the system will continue to test.

The alpha is in the matching pipeline and the asset class. A 10-rule structural quality filter added +\$10.73 in P&L with no model changes — direct empirical evidence that data quality dominates model choice at this scale. Oil near-expiry contracts produce a 76.5% backtest win rate while sports and politics are near-zero, because oil contracts converge mechanically on observable WTI-futures settlement.

The per-trade Sharpe of 0.501 (canonical) → 0.516 (leakage-free purged split) is the load-bearing headline. The Phase 18 audit (`AUDIT_REPORT.md`) confirmed the per-trade edge survives the embargo correction with only +2.99% drift, well within the bootstrap CI. The per-pair annualized Sharpe is regime-dependent (range 0.30–0.81 corrected, depending on cross-pair correlation) and reported with full caveats. An earlier draft cited per-pair annualized Sharpe of ≈ 3.2 ; the audit found no derivation path and replaced it with the leakage-free per-trade headline above.

Beyond the empirical findings, three broader lessons: (i) *infrastructure bugs masquerade as model problems* — Kalshi’s silent 429s, Polymarket’s `condition_id` typo, and our own `live_NNNN` schema drift caused weeks of confusion that looked like the model failing, and the fix in each case was a config change rather than a hyperparameter sweep; (ii) *evaluation regime matters more than model family* — a single train/test split can tell a very different story than a walk-forward, a per-category breakdown, or a data-scaling curve, so multi-regime evaluation is mandatory rather than optional; (iii) *negative results are results* — PPO failing

catastrophically is the strongest evidence we have for the main thesis, and we chose to publish it rather than omit it. A more defensible RL approach would require a curriculum that learns “safe” regimes first, a differentiable simulator for off-policy pre-training, and a much larger universe of training trajectories. None of these are justified at our data scale, but the negative result is itself informative for future practitioners.

Limitations. Our test window is short (two weeks of out-of-sample evaluation; the walk-forward analysis mitigates but does not eliminate this). Paper trading does not model market impact or partial fills. Survivorship bias is structural rather than retroactive (Phase 18 Tier 4 confirmed 0/10 retroactive drops in a random sample). Settlement-divergence risk is real but unobserved in our universe. The live commodity-matching pipeline was incomplete at submission and resolved post-submission, so the backtest oil edge has only a proof-of-life live validation. **Future work** includes 500- and 1000-bar scaling checkpoints (the auto-retrain system will produce these over the coming weeks), live-volume-aware microstructure features, a larger TFT configuration once data permits, and re-running the pre-registered LOGO feature ablation at 250+ bars/pair where bootstrap CIs become tight enough to classify individual feature groups as load-bearing or droppable.

The project succeeded on its own terms: we answered the research question rigorously, built a working autonomous system, and documented honest limitations. The edge is real; the complexity is not.

Acknowledgments

We thank Professor Kevin Gold for course instruction and for pivotal late-March feedback that pushed us to treat regression baselines as first-class and to add the PPO-without-autoencoder variant — feedback that directly shaped the research question into something empirically answerable. We acknowledge use of Anthropic’s Claude as an AI pair-programming assistant throughout implementation; all design decisions, experimental choices, and interpretations are our own.

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Appendix B: Figures

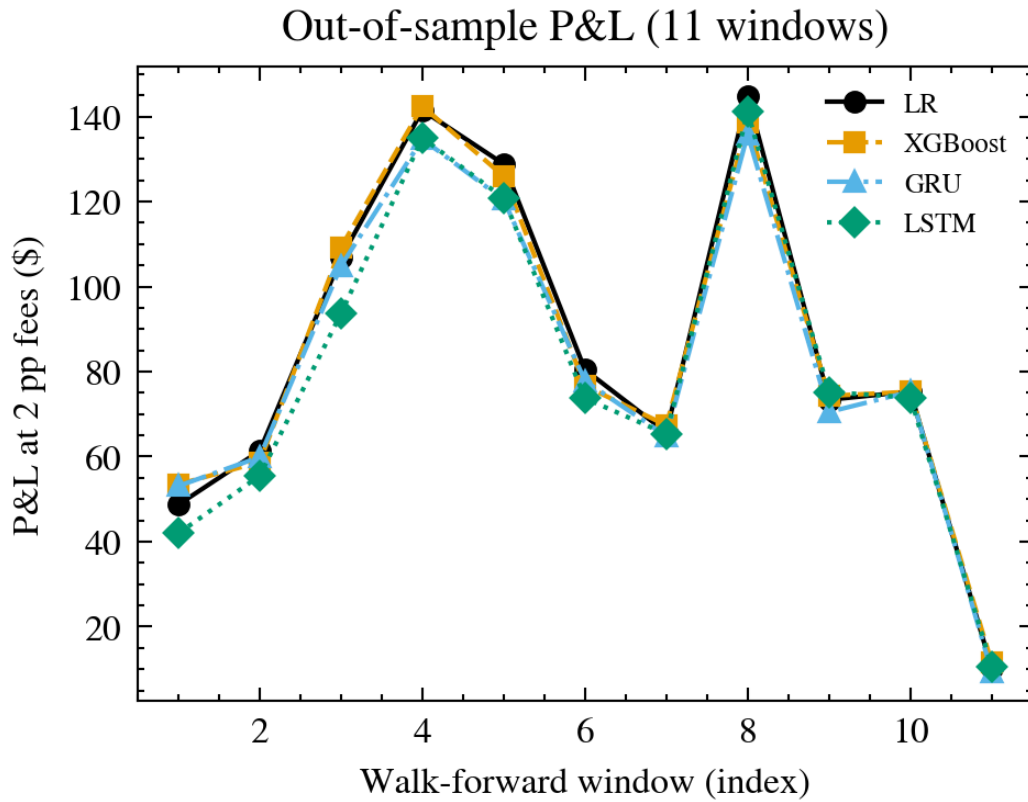


Figure 1. Out-of-sample P&L across 11 walk-forward windows (§3.2). Every ML model is profitable in every window; per-trade Sharpe rises from 0.31 in early windows to 0.53 in late windows, indicating the edge strengthens rather than decays as more data accumulates.

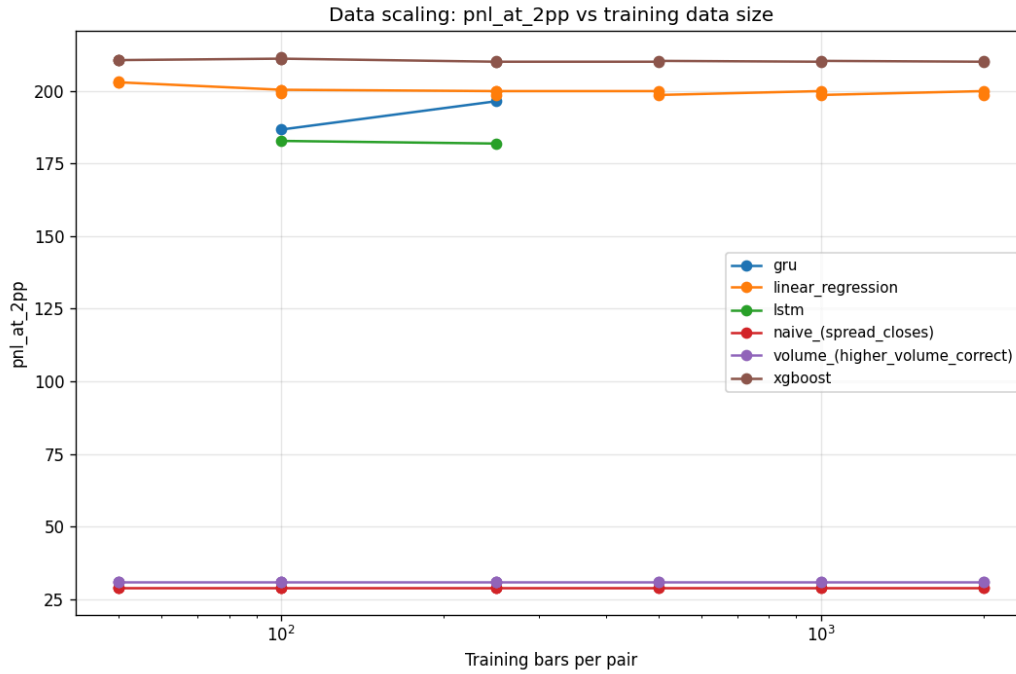


Figure 2. P&L vs. training-set size at 2 pp signal threshold (§3.4). The curve plateaus at $N=6,802$ because the underlying pair universe is fixed at 144 pairs with at most 141 bars each; slices at 250+ bars/pair are identical to the 100-bar slice. The plateau is a property of the fixed pair universe, not a universal scaling claim.

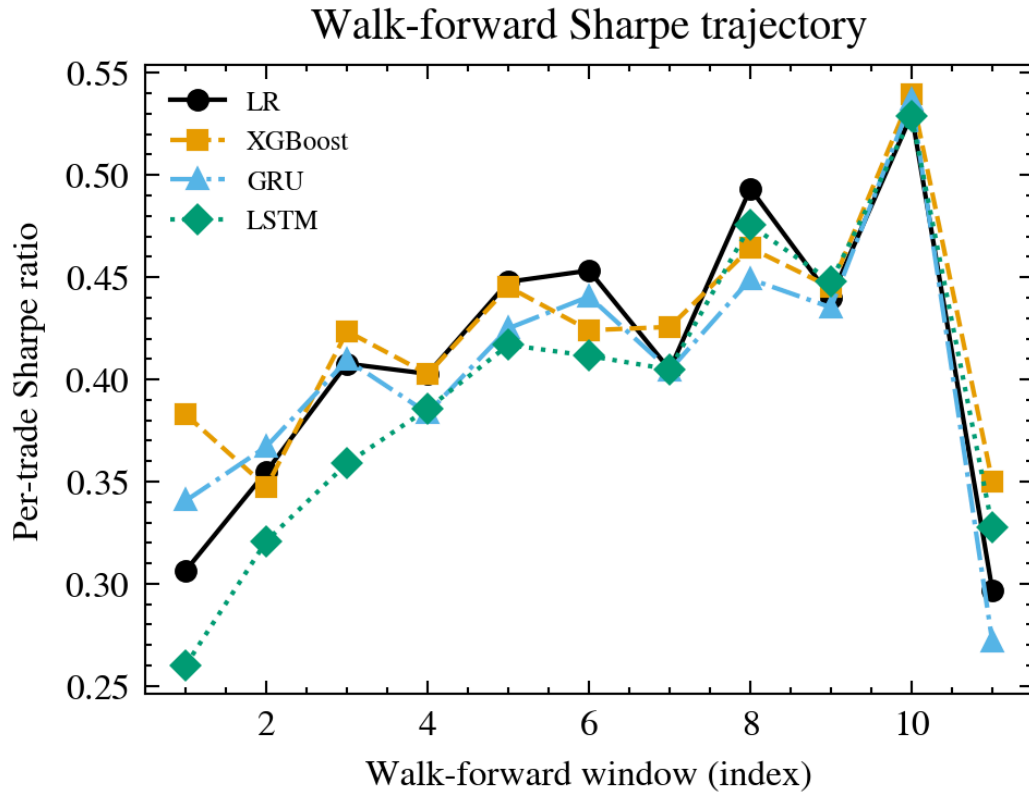


Figure 3. Walk-forward per-trade Sharpe trajectory across the 11 windows (§3.2 supplemental). The trend confirms the rising-edge pattern visible in Fig. 1.

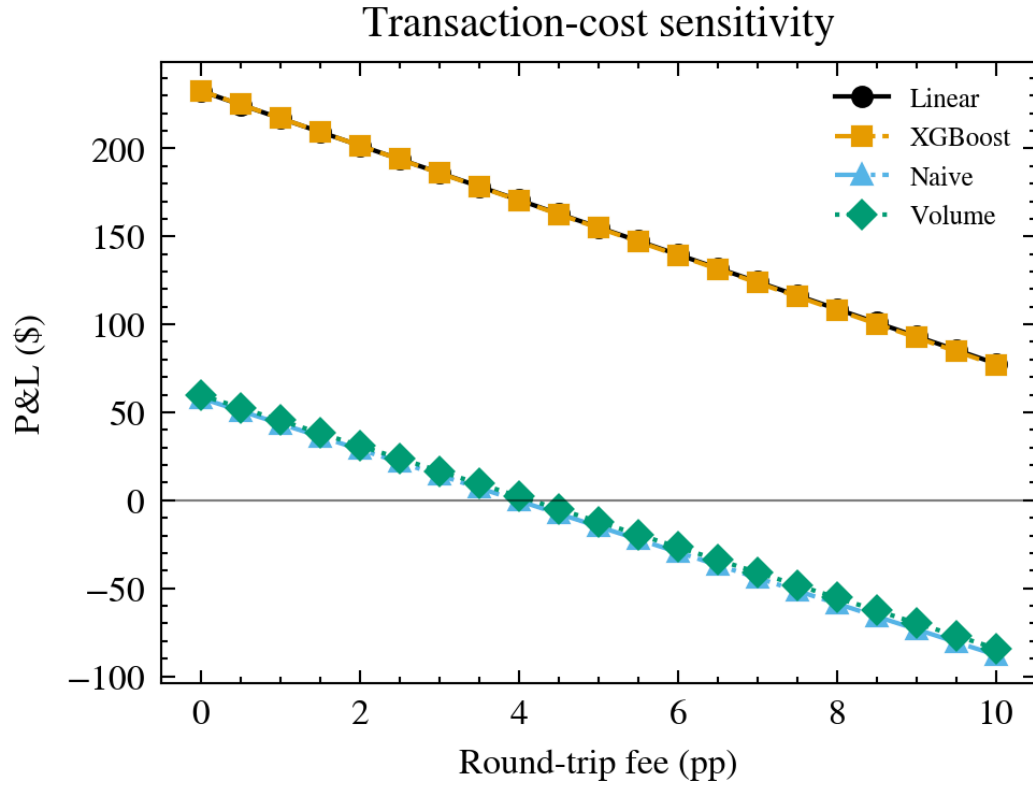


Figure 4. P&L vs. round-trip fee for the four ML models (§3.3). The rank-ordering of models is invariant to fee assumptions in the 0–10 pp range; LR and XGBoost remain the top tier across the entire fee spectrum.

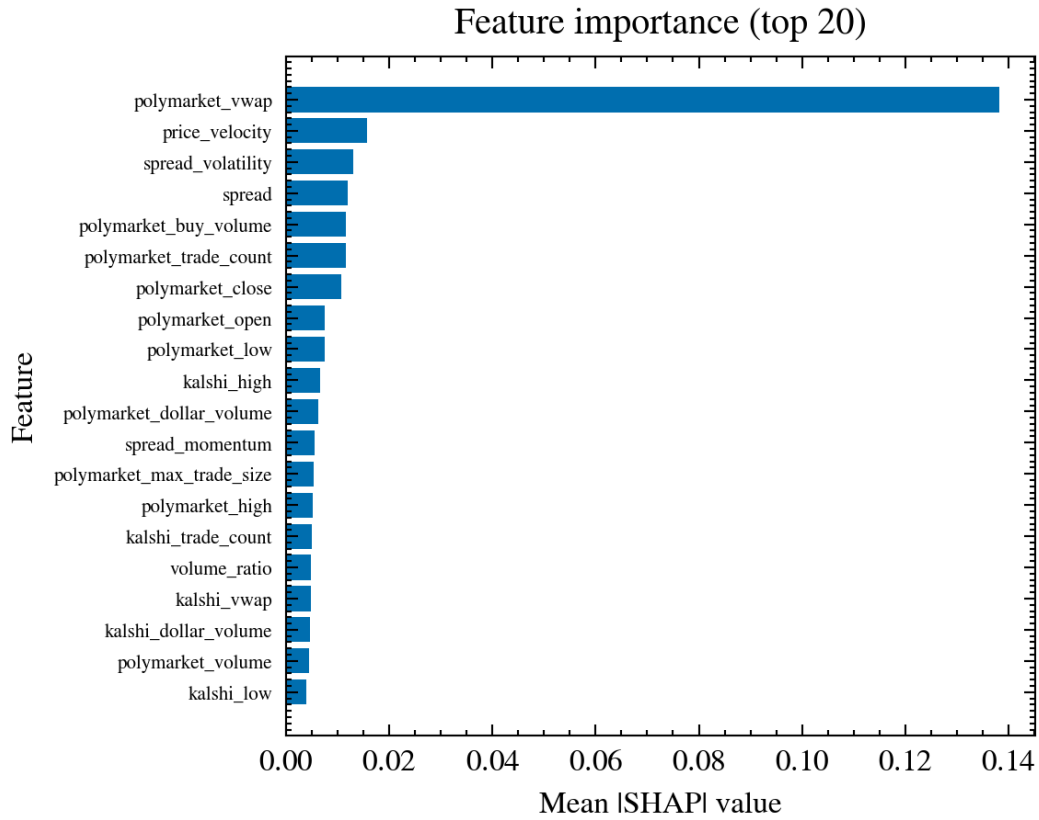


Figure 5. Mean |SHAP| feature importance for XGBoost (§3.4).

`polymarket_vwap` dominates with mean |SHAP| ≈ 0.14 , twice the next feature, suggesting Polymarket is the slower-reacting price discovery side.

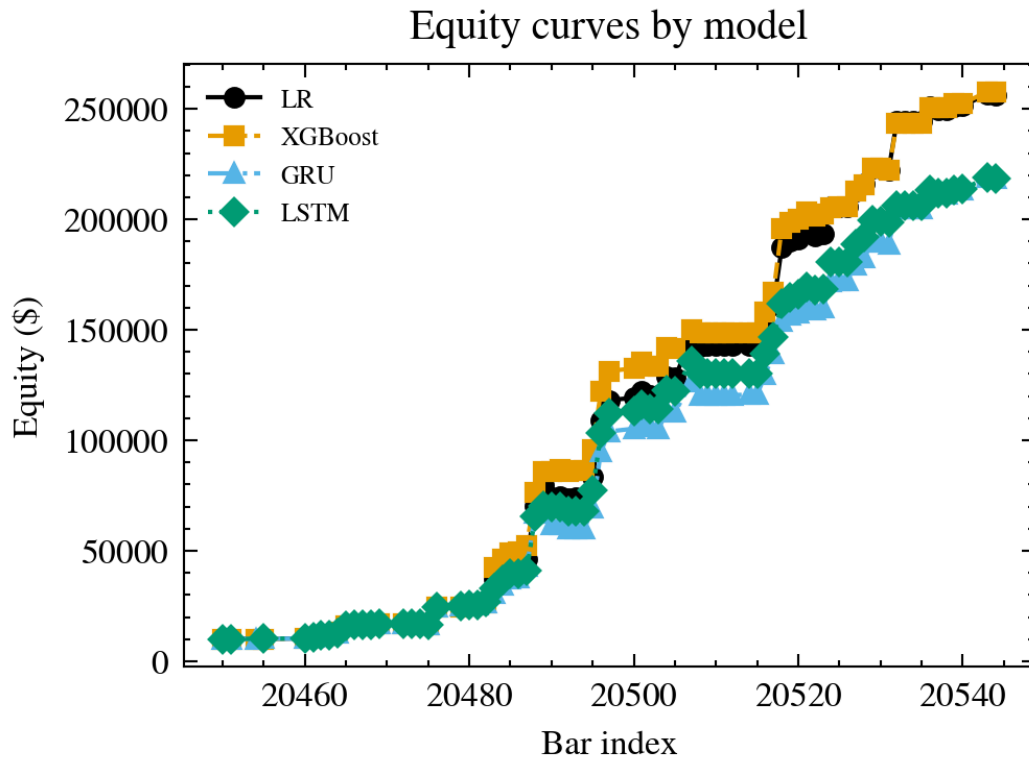


Figure 6. Cumulative test-set P&L by model (§3.1). The Tier-1 vs. Tier-2 separation opens gradually rather than in a single jump — consistent with a stable per-trade edge.

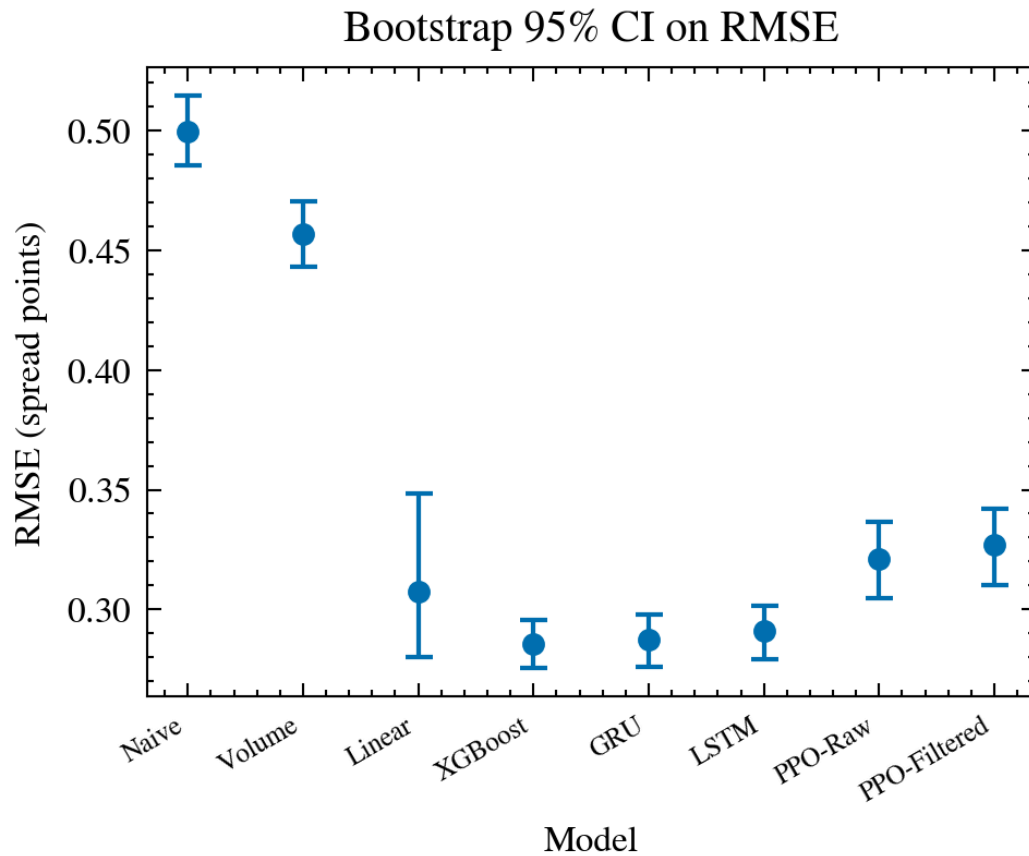


Figure 7. Bootstrap 95% CI on RMSE by model (1,000 resamples, §3.1). The LR / XGBoost / GRU / LSTM intervals overlap heavily even though P&L separates cleanly — confirming the P&L gap is driven by directional accuracy and trade selection, not raw regression error.

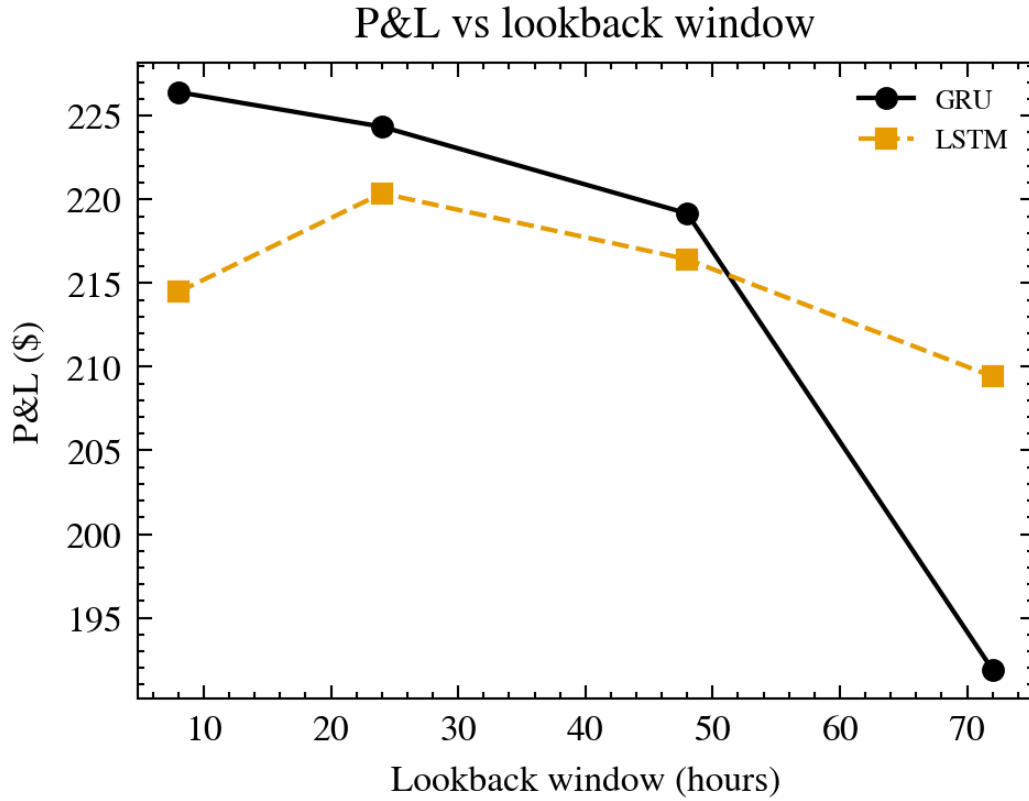


Figure 8. P&L vs. lookback window for GRU and LSTM (§3.3 / Proposal Experiment 2). Performance is essentially flat across {2, 6, 12, 18}-bar lookbacks, indicating the sequence models cannot exploit longer history at this dataset scale.

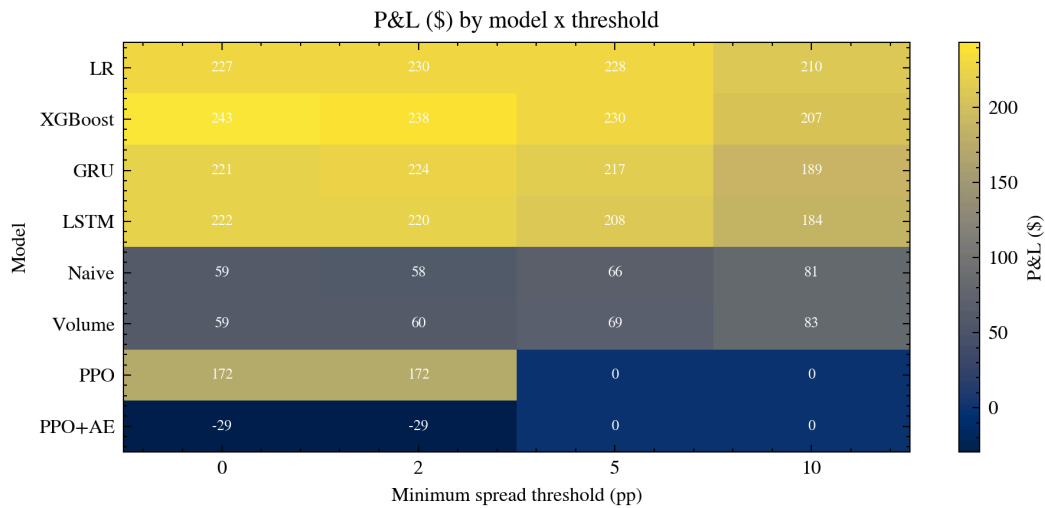


Figure 9. P&L heatmap by model $\times$ minimum-spread threshold (§3.3 / Proposal Experiment 3). Threshold = 2 pp dominates for all ML models; lower thresholds are dominated by noise, higher thresholds by missed trades.

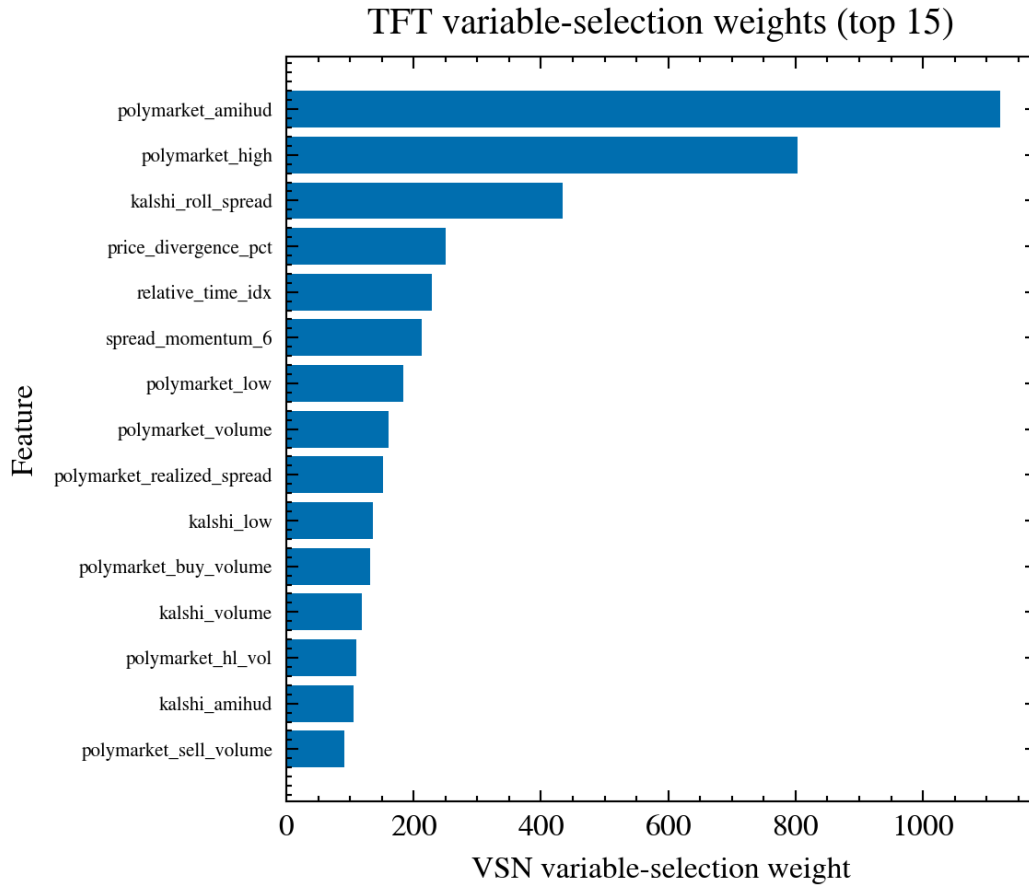


Figure 10. TFT VSN variable-selection weights (§3.4). The attention is healthy (entropy = 2.656 vs. degenerate baseline of ≈ 0) — TFT’s underperformance vs. GRU is a data-volume bottleneck at $N=6,802$, not an architectural failure.

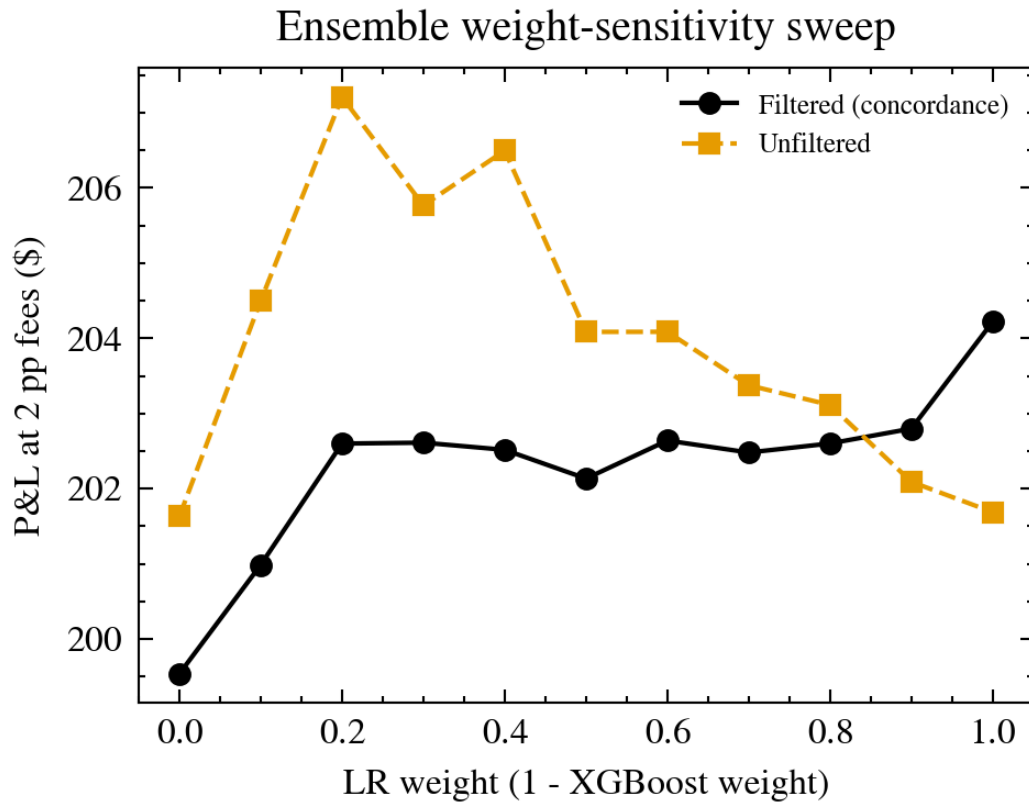


Figure 11. Walk-forward P&L at 2 pp fees across the 11-point LR-weight sweep (§3.4). The spread across weights is \$4.68; the curve is essentially flat, confirming the production ensemble’s weight choice is not cherry-picked.